

SOSVA, Russian Federation

WMO: 236250

Lat: 63.650N

Lon: 62.100E

Elev: 91

StdP: 14.65

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.5	(c) -37.5	(d) -47.3	(e) 0.3	(f) -42.3	(g) -42.0	(h) 0.5	(i) -37.0	(j) 10.9	(k) 16.8	(l) 9.5	(m) 14.5	(n) 0.5	(o) 120	(p) 0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	83.4	67.2	79.8	64.9	76.0	63.2	69.9	80.1	67.3	76.1	64.9	73.2	3.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	96.8	74.4	63.8	89.0	72.3	61.4	81.6	70.1	33.9	80.1	31.9	76.2	30.0	73.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.6	9.8	8.4	-43.2	88.2	7.4	2.8	-48.5	90.3	-52.9	91.9	-57.0	93.5	-62.4	95.6
			-43.1	72.7	6.8	2.9	-48.0	74.8	-51.9	76.5	-55.7	78.1	-60.7	80.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	-7.4	-2.8	15.3	29.5	43.0	56.1	63.0	56.0	45.7	31.8	9.0	-1.7	
	DBStd	27.11	16.18	16.00	12.71	10.60	9.70	9.04	7.73	7.18	7.28	9.73	17.57	18.50	
	HDD50	8847	1779	1477	1076	615	266	40	4	22	169	566	1229	1604	
	HDD65	13503	2244	1897	1541	1064	683	291	133	292	580	1030	1679	2069	
	CDD50	931	0	0	0	0	50	224	406	209	40	2	0	0	
	CDD65	110	0	0	0	0	1	24	71	14	0	0	0	0	
	CDH74	1183	0	0	0	0	49	254	748	129	3	0	0	0	
	CDH80	278	0	0	0	0	5	50	201	22	0	0	0	0	
Wind	WSAvg	2.9	2.2	2.5	3.3	3.6	3.7	3.4	2.6	2.8	3.0	3.1	2.6	2.4	
Precipitation	PrecAvg	18.4	1.0	0.8	0.9	1.1	1.5	2.5	2.4	2.5	2.0	1.5	1.3	0.9	
	PrecMax	25.1	2.0	1.5	2.4	2.8	4.6	4.8	6.2	5.4	4.4	3.3	3.1	1.7	
	PrecMin	11.2	0.2	0.0	0.0	0.0	0.2	0.8	0.3	0.7	0.7	0.4	0.0	0.4	
	PrecStd	3.7	0.5	0.4	0.6	0.6	1.0	1.2	1.7	1.2	1.0	0.7	0.7	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	34.5	45.3	58.2	79.8	85.5	89.0	82.8	71.3	57.9	39.0	35.1	
		MCWB	27.3	31.8	37.0	46.9	58.6	65.8	70.6	68.3	61.3	51.3	36.2	32.4	
	2%	DB	25.4	29.8	39.8	52.4	73.4	80.4	85.5	77.4	64.9	52.4	36.9	32.1	
		MCWB	24.4	27.6	34.2	42.5	56.3	63.7	68.5	65.8	57.0	47.1	34.5	29.5	
	5%	DB	22.0	24.8	36.6	47.6	66.7	76.5	82.1	72.9	60.7	48.1	34.2	28.0	
		MCWB	20.8	22.6	32.0	39.6	53.1	62.0	67.0	63.3	54.5	44.4	32.0	26.2	
	10%	DB	16.9	21.1	33.2	44.2	60.2	72.1	78.3	68.8	57.3	43.9	31.1	23.7	
		MCWB	15.6	19.3	29.5	37.6	49.3	60.1	64.8	60.8	52.3	41.2	29.3	22.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	32.1	37.8	47.4	61.9	68.9	73.6	71.4	64.9	52.2	36.9	32.8
			MCDB	28.2	34.4	43.4	56.8	73.2	78.9	83.4	80.4	68.2	57.1	38.3	34.7
2%		WB	24.4	27.8	34.7	43.0	57.0	66.2	71.1	67.6	57.9	47.6	34.7	29.5	
		MCDB	25.2	29.9	39.1	51.0	69.5	76.7	82.1	74.7	63.9	51.7	36.6	32.0	
5%		WB	20.8	22.8	32.2	40.4	53.5	63.8	69.0	64.5	54.9	44.4	32.2	26.3	
		MCDB	22.0	24.5	36.2	46.3	65.9	73.7	79.0	70.8	59.5	47.9	34.1	27.9	
10%		WB	15.7	19.2	29.8	38.3	50.1	61.1	66.6	61.8	52.7	41.4	29.2	22.5	
		MCDB	17.0	21.2	33.2	43.4	59.3	70.3	75.1	68.0	56.9	43.7	31.0	23.8	
Mean Daily Temperature Range	5% DB	MDBR	14.8	17.4	19.8	18.0	17.8	17.4	19.3	17.3	13.8	11.3	12.6	14.5	
		MCDBR	14.0	16.7	17.9	21.5	28.6	23.5	24.7	23.0	19.1	14.7	11.1	15.3	
		MCWBR	13.3	15.0	13.5	14.1	14.8	10.8	10.9	12.8	11.9	10.7	9.9	13.8	
	5% WB	MCDBR	13.9	15.5	17.5	18.4	26.2	21.2	21.3	19.8	17.6	14.5	10.6	14.9	
		MCWBR	13.3	14.4	13.6	12.6	14.3	11.5	11.4	12.6	11.9	10.9	9.8	13.8	
Clear-Sky Solar Irradiance	taub	0.226	0.283	0.324	0.356	0.362	0.374	0.417	0.386	0.339	0.302	0.226	0.225		
	taud	1.934	2.045	2.049	2.110	2.361	2.417	2.284	2.354	2.455	2.345	1.992	1.914		
	Ebn at Noon	109	194	235	257	270	268	250	248	239	196	108	42		
	Edh at Noon	10	22	33	39	33	32	36	31	23	16	9	5		
All-Sky Solar Radiation	RadAvg	63	299	769	1275	1610	1674	1697	1165	642	292	83	25		
	RadStd	3	23	56	140	134	128	146	121	85	33	11	1		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page